

Government College University Faisalabad

Department of Software Engineering
BS in Software Engineering

Final Year Project Proposal

Project Title:

Smart Hotel Website Auto-Generator

Course Code:

Final year project

Session:

2022 – 2026

Student Information:

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Date of Submission:

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1. Problem Statement

Small and medium-scale hotels in Pakistan face serious difficulties in developing and maintaining professional websites. Custom website development is expensive, time-consuming, and requires technical expertise that most hotel owners do not have. Hiring developers can cost between Rs. 200,000 to Rs. 500,000 and may take several months.

Additionally, even small changes in the website require technical support, which further increases cost and dependency. There is no standard automated solution that allows hotel owners to independently create and manage a complete hotel website according to their needs.

2. Background Analysis

With the rapid growth of online booking platforms and digital services, hotels are expected to maintain an online presence. However, most existing solutions are either costly SaaS platforms with monthly fees or require full-time developers.

Modern web technologies such as Next.js, React, and automated code generation have made it possible to build systems that generate complete applications automatically. Despite this, no affordable and customizable auto-generation system exists specifically for hotel websites in the local market.

3. Project Rationale / Motivation

The motivation behind this project is to reduce cost, time, and technical dependency for hotel owners. The system aims to empower non-technical users to generate their own professional hotel website within minutes.

This project also demonstrates modern software engineering practices such as automation, modular architecture, and scalable system design, making it a strong academic and practical contribution.

4. Developer and Target Users Aim and Objectives

Developer Objectives

- To design an automated website generation system
- To implement modular architecture for flexibility
- To demonstrate modern web development skills
- To build a scalable and reusable system

Target Users

- Small and medium hotel owners
- Guest houses and resorts
- Non-technical business owners

System Objectives

- Allow users to select required modules
- Automatically generate a complete working website
- Provide admin and customer dashboards
- Ensure mobile responsiveness and usability

5. Scope and Limitation of the Project

Scope

- Website auto-generation for hotels
- Module-based system (Rooms, Booking, Services, Billing, etc.)
- Admin and customer dashboards
- Downloadable source code
- MongoDB database integration

Limitations

- Online payment gateways not included in Phase-1
- Mobile app version not included
- Limited to hotel domain only
- Requires internet for generation process

6. Proposed Software Models

The **Agile Development Model** will be used for this project.

- Iterative development
- Continuous testing
- Module-by-module implementation
- Easy handling of changes

Prototype approach will also be used for UI validation.

7. System Design (Conceptual Overview)

- User Authentication System
- Module Selection Dashboard
- Code Generation Engine
- Template Library
- Database Schema Generator
- ZIP File Export System

System will follow **MVC-based modular architecture**.

8. Expected Outcomes / Anticipated Framework

- Fully functional auto-generated hotel website
- Admin dashboard for hotel management
- Customer booking and services system
- Modular and scalable architecture
- Clean UI with responsive design
- Ready-to-deploy project structure

9. Project Timeline (Gantt Chart – Summary)

Week	Activity
1–2	Requirement analysis & UI design
3–4	Module template development
5–6	Code generation engine
7	Services module integration
8	Dashboard & main system
9	Testing & debugging
10	Deployment & documentation

10. Tools / Resources / AI Resources

Development Tools

- Next.js/React js
- Tailwind CSS
- Node.js
- MongoDB/postgreSQL
- Prisma ORM

Other Tools

- Git & GitHub
- VS Code
- Figma

AI Resources

- ChatGPT (code assistance & documentation support)

11. Risk Assessment and Mitigation Plan

Risk	Mitigation
Module integration issues	Incremental testing
Time constraints	Agile planning
Code complexity	Modular design
Deployment errors	Multiple test environments

12. Similar Successful Projects (Optional)

- Website builders like Wix and WordPress
- SaaS hotel management systems
- Code generation frameworks

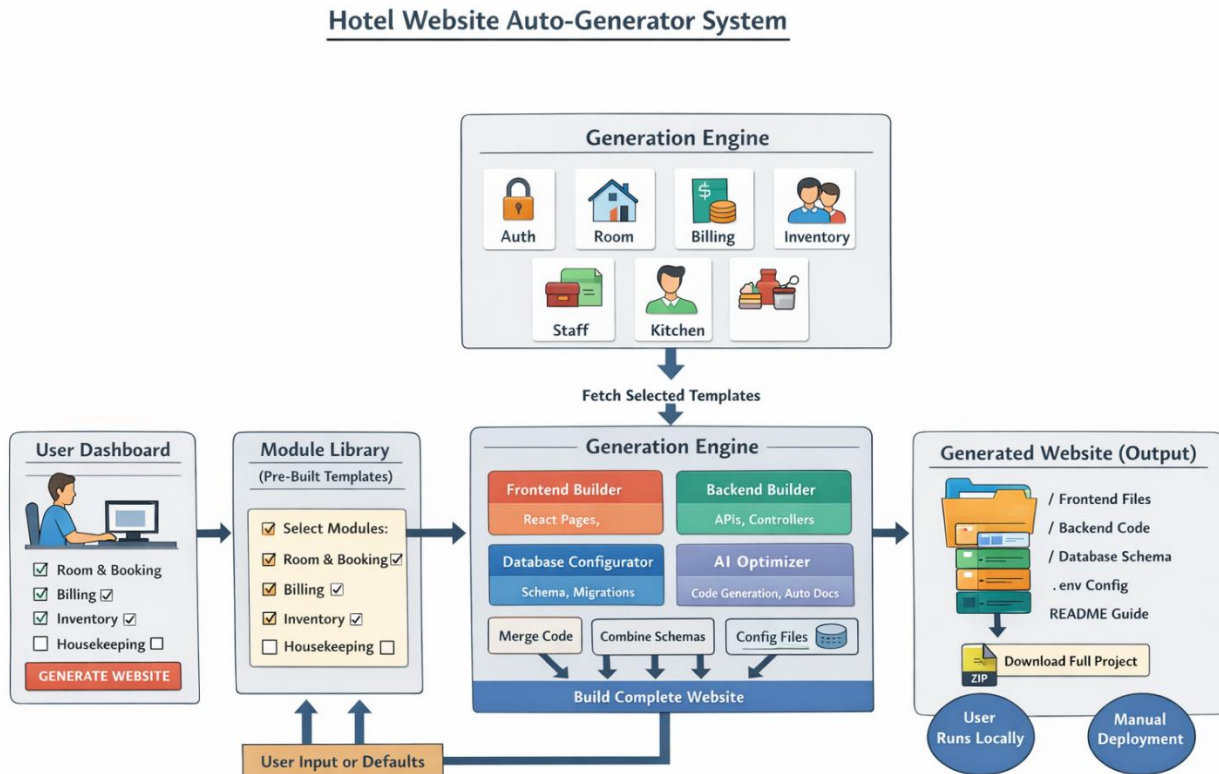
However, these platforms do not provide **full downloadable source code** like the proposed system.

13. Supervisor’s Remarks

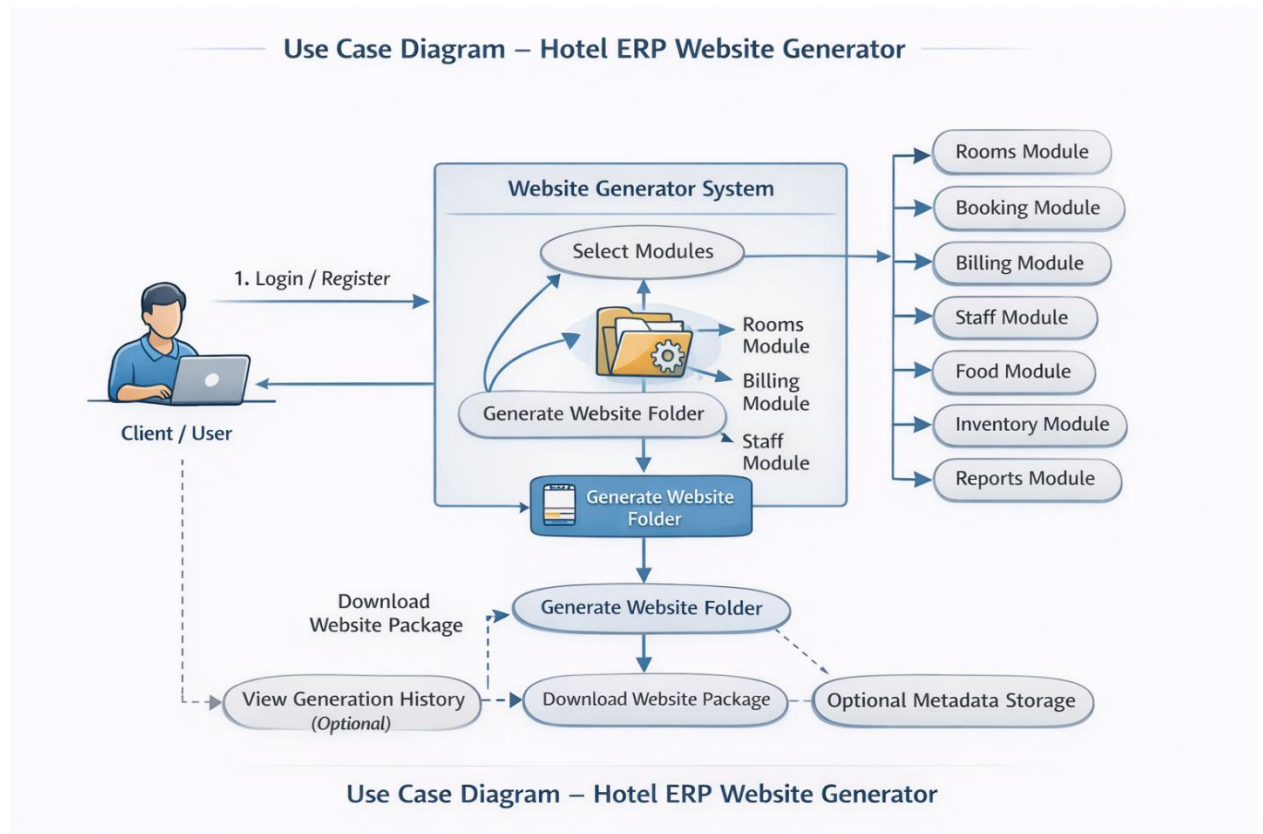
Appendices

Appendix A

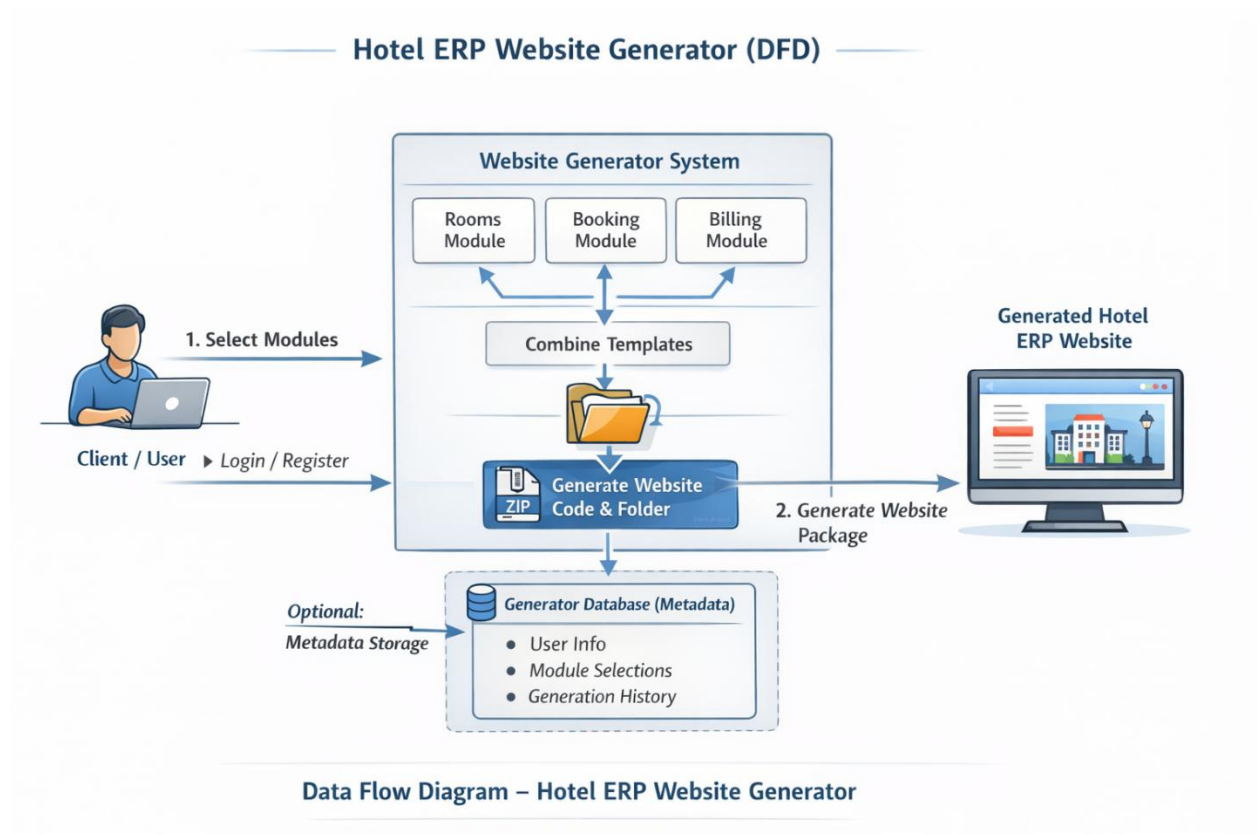
- System Architecture Diagram



- **Use Case Diagram**



- **DFD (Data Flow Diagram)**



Appendix B

- UI Mockups (Dashboard, Booking Page, Services Page)

Admin Dashboard:



Grand Luxury Hotel

Admin Dashboard



Admin



Total Rooms

32 Available

45



Active Bookings

+5 Today

28



Today's Revenue

+12% vs yesterday

₹45K



Total Guests

15 Check-ins today

156

Activate Windows

Go to Settings to activate Windows.

Recent Bookings

GUEST	ROOM	CHECK-IN	STATUS	ACTION
Ahmed Ali	Deluxe 101	Dec 22, 2024	confirmed	View
Sara Khan	Suite 205	Dec 23, 2024	confirmed	View
Bilal Ahmed	Standard 103	Dec 22, 2024	pending	View
Fatima Hassan	Deluxe 204	Dec 24, 2024	confirmed	View

Appendix C – Module Description Table

Module Name	Description	Key Features / Responsibilities	Notes
Rooms / Room Management	Manage all hotel rooms, types, and status	Add / Edit / Remove rooms, Track room type (Single, Double, Suite), Room availability status	Core module for bookings
Booking / Reservation	Handle customer room reservations	Create / Update / Cancel bookings, Track check-in/check-out dates, Availability validation	Integrates with Rooms module
Customer / Guest Management	Manage customer information	Store guest personal info, Maintain history of stays, VIP / Regular categorization	Connects with Booking & Billing
Billing & Invoice	Generate bills for customers	Automatic invoice generation, Payment tracking, Discounts & taxes calculation	Connects with Booking & Services modules
Housekeeping / Services	Track housekeeping and hotel services	Room cleaning status, Laundry / amenities tracking, Service requests	Integrates with Rooms module
Inventory / Stock Management	Manage hotel inventory	Track food, consumables, and housekeeping stock, Low-stock alerts, Update stock usage	Connects with Services / Kitchen modules
Staff / Employee Management	Manage hotel staff	Employee profiles, Roles & permissions, Attendance tracking	Required for admin and operational control
Food Ordering / Kitchen System	Manage restaurant and room service	Menu management, Order tracking (room & restaurant), Billing integration	Connects with Inventory & Billing
Reports / Analytics	Generate reports for admin	Occupancy & revenue reports, Staff performance, Inventory usage stats	Optional for dashboard & management review
Authentication / User Management	Secure login for admin, staff, customers	Register / Login / Logout, Role-based access control, Password reset	Core for all other modules